

INTEGRATED ANTENNA 1U CUBESAT EXPERIMENT WITH MULTI-NEEDLE LANGMUIR PROBES IN THE GHOST STUDENT MISSION

Asad Jonas Munir Arshad (1)

(1) Lyra / University of Oslo, Department of Physics, Sem Sælands vei 24, 0371 Oslo, Norway, Email: styreleder@lyrafutures.org / ajarshad@uio.no

ABSTRACT

Munir-3 was a 1U CubeSat experiment combining a frame-integrated antenna with a four-channel multi-Needle Langmuir Probe. No telemetry was received during flight, so spacecraft release, activation, RF operation and probe deployment could not be verified. Post-flight RF tests demonstrated valid packet reception over approximately 125–700 m, while no usable link was established at approximately 4.07 km. Plasma-chamber tests showed that restoring 114.75 cm² of known spacecraft-ground-bonded copper relative to a reduced-area configuration made the powered floating spacecraft 0.59–0.84 V less negative. The results demonstrate that structural antenna integration, conductive spacecraft area and fixed-bias plasma diagnostics must be treated as coupled spacecraft-level design parameters.

1. INTRODUCTION

Small spacecraft increasingly employ multifunctional structures to reduce mechanisms, interfaces and occupied volume. Integrating an antenna into the spacecraft structure is one example, but doing so couples radio-frequency (RF), mechanical and electrical design decisions that are often treated separately. This coupling becomes especially important when the spacecraft also carries fixed-bias plasma instrumentation, because probe potentials are referenced to spacecraft electrical ground while collected electron current must be balanced by currents exchanged through electrically connected spacecraft surfaces [1].

Munir-3 was developed by Lyra as a one-unit (1U) CubeSat experiment for the Grand cHallenge MesOsphere Student rocket (GHOST) mission.

The spacecraft combined a frame-integrated antenna with a four-channel multi-Needle Langmuir Probe (mNLP). The antenna concept favoured an electrically insulating structural frame, whereas the mNLP required conductive spacecraft surfaces connected to electrical ground to participate in plasma-current closure. The resulting architecture therefore linked structural material selection, antenna integration, spacecraft charging and plasma diagnostics at system level.

Munir-3 flew on the GHOST sounding-rocket mission launched from Andøya on 18 November 2025. No telemetry was received from Munir-3, so successful spacecraft release, activation, RF operation and in-flight mNLP deployment could not be independently verified [2]. The investigation therefore combined pre-flight verification records with controlled post-flight RF and plasma-chamber tests.

This paper presents the implemented spacecraft architecture, the principal pre-flight verification results, and the post-flight investigation of the RF link and mNLP operating conditions. Particular attention is given to how the electrically insulating antenna structure and spacecraft-grounded conductive exterior affected the operating point of the fixed-bias plasma diagnostic. The objective is not to assign a unique cause to the missing telemetry, but to establish the demonstrated system-level interactions and the limits of the available evidence.

2. MUNIR-3 SYSTEM DESIGN

2.1 Frame-integrated antenna and spacecraft architecture

Munir-3 used Ketron™ GF30 PEEK, an extruded PEEK containing 30% glass-fibre reinforcement,

for the principal load-carrying structural components [3]. The material provided an electrically insulating frame into which the antenna conductor could be integrated without making the complete structure part of spacecraft electrical ground [2].

In the flight spacecraft, the antenna conductor followed the recessed inner perimeter of the structural frame and connected through a frame-mounted SMA interface. The arrangement eliminated a separate post-ejection antenna-deployment mechanism and mechanically integrated the antenna with the 1U structure [2].



Figure 1. *Munir-3 flight hardware and frame-integrated antenna architecture. (a) GF30-PEEK structural frame with recessed antenna conductor and frame-mounted SMA feed. (b) Assembled 1U spacecraft showing the exterior PCB panels and integrated structural configuration.*

The communications subsystem used a Texas Instruments CC1200 transceiver with a final flight carrier of 436 MHz. Flight software used High Performance mode, transmit-power ramping and the maximum software transmit-power setting, with a project working value of approximately +14 dBm. This is a configured CC1200 value rather than a measured antenna-port or radiated output. Final antenna gain, efficiency, matching and post-amplifier output were not independently established [2].

The exterior spacecraft assembly included two side dummy PCBs, a top dummy PCB and the bottom/deployment PCB. Exposed copper regions on these surfaces were electrically bonded to spacecraft ground. Four quantified exterior copper regions totalled 114.75 cm² and formed part of the flight-as-built spacecraft conductive exterior [2].

2.2 Multi-Needle Langmuir Probe

The mNLP used four fixed-bias cylindrical probes with nominal spacecraft-referenced biases of

$$P_1 = 9 \text{ V}, \quad P_2 = 5 \text{ V}, \quad P_3 = 8 \text{ V}, \quad P_4 = 6 \text{ V}. \quad (1)$$

Each probe assembly was 80.8 mm long and contained a 44.0 mm biased guard section followed by a 33.8 mm exposed centre-conductor collector. The collector diameter was 0.51 mm, giving an active cylindrical area

$$A_p = \pi dL \approx 5.42 \times 10^{-5} \text{ m}^2. \quad (2)$$

Two antiparallel probes were mounted on each of two payload PCBs. Each pair shared one burn-wire restraint, while each probe retained its own spring and MCX connection [2].

The fixed-bias mNLP estimates electron density from simultaneous electron-current measurements at several positive biases. Under cylindrical orbital-motion-limited (OML) conditions,

$$M = \frac{dI_e^2}{dV_b}, \quad (3)$$

and the density inversion used here is

$$n_e = \frac{\sqrt{M}}{CA_p}, \quad C = \frac{e^{3/2}}{\pi} \sqrt{\frac{2}{m_e}}, \quad (4)$$

where I_e is collected electron current and V_b is probe bias [4].

Because the probe biases are referenced to spacecraft ground, spacecraft charging shifts all four probes together relative to the surrounding plasma. A sufficiently negative spacecraft potential

can therefore move lower-biased probes away from the intended electron-collection regime even though the relative spacing between the four biases remains unchanged [1].

The 0.51 mm collector diameter was small compared with the estimated chamber Debye length, but the 33.8 mm active length was comparable to the estimated 34–39 mm Debye length. The ideal long-cylinder condition was therefore not satisfied. Values reconstructed from the four-point relation are consequently reported as OML-derived apparent electron densities, not independently validated absolute plasma densities.

2.3 Conductive-area and electrical architecture

For an electrically floating spacecraft, equilibrium potential is determined by the net current exchanged through the electrically connected spacecraft and payload surfaces. Positive mNLP probe currents therefore require corresponding return-current paths. Munir-3 combined an electrically insulating PEEK frame with spacecraft-grounded copper exterior surfaces specifically relevant to this current balance [1].

The 114.75 cm² quantified copper area was part of the flight spacecraft. During the post-flight chamber campaign these exterior panels were physically removed to create a reduced-area configuration and then restored to reproduce a flight-like exterior-panel configuration [2].

The spacecraft used three Panasonic NCR18650GA Li-ion cells with a common negative return. Two cells supplied the general electronics branch, while a separate positive feed from the third cell powered the burn-wire circuit. The EPS generated nominal 3.3, 5, 10 and 12 V rails. Firmware implemented a 15 s safety delay followed by 15 s burn-wire activation before normal measurement operation. Ground tests produced probe release within approximately 3–9 s of energization; this verifies the mechanism only under the tested ground conditions [2].

3. METHODS AND VERIFICATION

3.1 Pre-flight verification

Standalone mechanical verification was performed at Kongsberg Defence & Aerospace before

launcher integration at NASA Wallops Flight Facility (WFF). The campaign included low-level resonance surveys, 10 g peak sine excitation and 12.7 g_{rms} random vibration. The assembled spacecraft met the project-level survival criterion with structure and electronics remaining intact and functional. Local X- and Z-axis “out of spec” annotations remain in the detailed resonance records but are not interpreted as spacecraft-level failure. Launcher-integrated vibration testing was subsequently performed at WFF; no quantitative executed WFF vibration record is available in the present source set [2].

Electrical verification included continuity checks, nominal rail verification and integrated end-to-end operation. A recorded running-state input current of 0.55 ± 0.02 A was measured at the simulated battery input, and ground testing demonstrated data acquisition, processing and transmission to a dummy ground station. These tests demonstrate functionality under the tested conditions but do not establish worst-case combined-load or flight electrical margin [2].

Development testing identified faults in earlier 5 V and 12 V implementations that were corrected before flight. The final EPS nevertheless retained two complete 12 V converter implementations connected to the same output rail. Separate ground testing also documented intermittent overcurrent events that persisted after external subsystem PCBs were disconnected. These findings are treated as pre-flight reliability evidence; neither their component-level cause nor their occurrence in flight was established [2].

3.2 Post-flight RF tests

Following the absence of flight telemetry, the 436 MHz communication chain was tested using two post-flight 2-GFSK configurations: 40 kbit/s with 25 kHz deviation and 208 kHz receiver bandwidth, and 1.2 kbit/s with 4 kHz deviation and 11 kHz receiver bandwidth.

Dedicated short-range tests were performed separately over approximately 125–700 m under approximately line-of-sight conditions. Receiver logs were used to identify successful packets and

reception errors. Separate long-range tests were then performed at approximately 4.07 km [2].

Free-space path loss was calculated only as a geometric reference,

$$L_{FS}=20\log_{10}\left(\frac{4\pi R}{\lambda}\right), \quad (5)$$

giving approximately 82.1 dB at 700 m and 97.4 dB at 4.07 km for 436 MHz [5]. These values are not treated as a closed link budget because radiated power, antenna performance and the complete receive chain were not independently established.

3.3 Plasma-chamber method and analysis

Post-flight plasma testing was performed in an argon plasma chamber using a rotating spacecraft fixture. The principal analysis used nominal 3 SCCM operation, with additional matched floating-spacecraft measurements at 2.25 and 1.5 SCCM.

In the floating condition, spacecraft ground was not intentionally hard-connected to the chamber wall. In the grounded control, spacecraft ground was connected to the chamber wall, constraining the spacecraft reference close to Earth ground. Spacecraft potential relative to chamber/Earth ground was measured with a UNI-T UT161D digital multimeter having approximately 10 MΩ input impedance; the floating condition therefore denotes absence of an intentional low-impedance ground connection rather than perfect electrical isolation.

The principal conductive-area comparison used a reduced-area chamber configuration, with the four flight exterior PCB panels absent, and a flight-like exterior-panel configuration, with those panels restored and their 114.75 cm^2 of quantified exposed copper bonded to spacecraft ground. This value is an increment relative to the reduced-area chamber configuration and is not the total conductive spacecraft area [2].

The four mNLP channels were sampled internally at 250 Hz per probe, while the chamber dataset stored one complete probe-and-attitude record

every 0.1 s. Post-flight end-to-end calibration was performed with a precision source/measure unit. The publication analysis was restricted to the measured 0.1–4.0 μA interval [2].

Quantitative OML reconstruction was restricted to grounded Exp. 3.1.3 and Exp. 5.1. At each timestamp, calibrated probe currents were squared and fitted against the nominal 5, 6, 8 and 9 V biases. Only fits satisfying

$$R^2 \geq 0.95 \quad (6)$$

were retained. This threshold is a linearity filter only; it is not an accuracy criterion and does not demonstrate satisfaction of ideal cylindrical-OML assumptions [6].

4. RESULTS

4.1 Pre-flight verification and flight outcome

The assembled Munir-3 spacecraft met the project-level survival criterion during the standalone Kongsberg Defence & Aerospace vibration campaign, with the structure and electronics remaining intact and functional. Local X- and Z-axis response annotations did not correspond to an observed spacecraft-level failure. Launcher-integrated vibration testing was subsequently completed at NASA Wallops Flight Facility, but no quantitative executed WFF record is available.

Pre-flight electrical testing demonstrated operation of the nominal power rails and successful end-to-end data acquisition, onboard processing and transmission to a dummy ground station. A spacecraft running-state input current of 0.55 ± 0.02 A was measured under one recorded ground-test condition. Separate ground testing also documented an intermittent EPS-side overcurrent that reached the 1.5 A bench-supply current limit and persisted after external subsystem PCBs were disconnected; its component-level cause was not established [2].

During flight, no telemetry was received from Munir-3. Because the CubeSat Eject Pod provided no independent release-status or spacecraft

electrical interface, successful ejection, activation, RF operation and mNLP deployment remain unverified. The flight observation is therefore the absence of telemetry, not evidence for a particular failure mechanism [2].

4.2 Post-flight RF investigation

Both post-flight 2-GFSK configurations produced valid packet receptions during the approximately 125–700 m line-of-sight tests, while neither established a usable link at approximately 4.07 km. The 40 kbit/s long-range record contained repeated RX_TIMEOUT events and no successful RX_OK reception; no receiver logfile was preserved for the corresponding 1.2 kbit/s test [2].

Because final antenna performance, post-amplifier output and the complete receiving chain were not independently established, the RF tests do not constitute a closed link budget. They demonstrate short-range post-flight functionality and failure to establish the tested long-range links, but do not identify the frame-integrated antenna or modem configuration as the controlling limitation [2].

4.3 Plasma-chamber investigation

The strongest chamber result was obtained from the matched powered-floating comparison between the reduced-area chamber configuration and the flight-like exterior-panel configuration. Probe geometry and powered state were held constant while the four flight exterior PCB panels, corresponding to 114.75 cm² of known exposed spacecraft-ground-bonded copper, were restored.

Table 1. *Powered floating-spacecraft potential relative to chamber/Earth ground for the matched conductive-area comparison [2].*

Nominal Ar flow [SCCM]	Reduced-area configuration [V]	Flight-like exterior panels [V]	Change [V]
3.00	-10.58	-9.89	+0.69
2.25	-10.91	-10.32	+0.59
1.50	-11.03	-10.19	+0.84

Restoring the flight exterior panels made the powered floating spacecraft 0.59–0.84 V less negative relative to chamber/Earth ground across

all three matched flow conditions. Because plasma potential was not independently measured, the shifts are not interpreted as motion toward plasma potential [2].

The chamber-grounded controls remained close to the chamber/Earth-ground reference, at +0.054 V for Exp. 3.1.3 and +0.092 V for Exp. 5.1. At 3 SCCM, grounding shifted the spacecraft potential by +10.63 V and +9.98 V for the reduced-area and flight-like configurations, respectively, demonstrating the substantially larger effect of a low-impedance external return path.

Spacecraft power state also affected the floating equilibrium: powering the spacecraft made its potential more negative at every tested flow, while the power-on excursion was 0.88–1.00 V smaller with the flight exterior panels present. Probe-orientation dependence further showed that biased-probe collection contributed to the current balance [2].

The rotating grounded datasets showed that four-point OML fit quality was strongly geometry dependent. Only 2 of 1584 timestamps in Exp. 3.1.3 and 41 of 1488 timestamps in Exp. 5.1 satisfied the adopted $R^2 \geq 0.95$ linearity criterion [2].

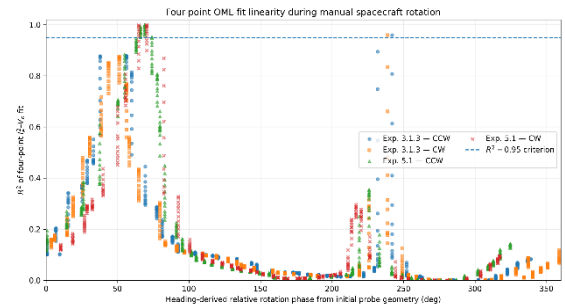


Figure 2. *Four-point I_e^2 -versus- V_b fit linearity during post-flight manual spacecraft rotation for grounded Exp. 3.1.3 and Exp. 5.1 at nominal 3 SCCM. Relative phase is heading-derived. The dashed $R^2=0.95$ line is the analysis-selection criterion and is not an accuracy threshold [6].*

For the retained timestamps, Exp. 3.1.3 produced an OML-derived apparent electron density of approximately

$$8.25 \times 10^{10} \text{ m}^{-3}, \quad (7)$$

while Exp. 5.1 produced a median apparent density of approximately

$$1.80 \times 10^{11} \text{ m}^{-3} \quad (8)$$

[2]. The corresponding four-point relations are shown in Fig. 3.

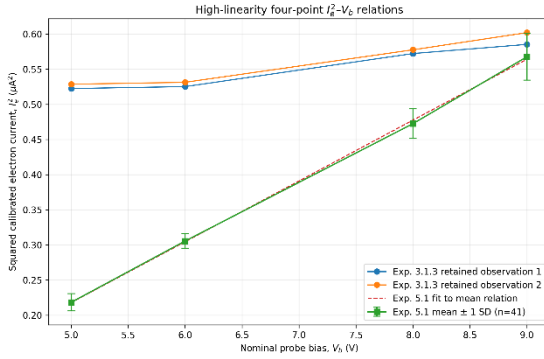


Figure 3. High-linearity four-point I_e^2 -versus- V_b relations for grounded Exp. 3.1.3 and Exp. 5.1. The two retained Exp. 3.1.3 timestamps are shown individually; Exp. 5.1 markers represent the mean of the 41 retained samples with sample-to-sample variation. The different slopes correspond to different OML-derived apparent-density estimates but do not establish absolute density accuracy.

The larger retained fraction in Exp. 5.1 indicates more frequent agreement with the four-point linear $I_e^2 - V_b$ relation, not greater absolute accuracy. Because the collector length was comparable to the estimated Debye length and plasma potential was not independently measured, the reconstructed values remain OML-derived apparent densities.

5. DISCUSSION

Munir-3 demonstrates that structural antenna integration and fixed-bias plasma instrumentation cannot be treated as independent subsystem problems. The electrically insulating GF30-PEEK frame enabled the antenna conductor to be incorporated into the load-carrying structure, while the mNLP depended on spacecraft-grounded conductive surfaces to provide plasma-current closure [1]. The chamber measurements show that this distinction was operationally significant.

The clearest result is the matched powered-floating comparison in which restoring the flight exterior panels and their 114.75 cm² of known ground-bonded copper made the spacecraft potential 0.59–0.84 V less negative across all three tested argon-flow conditions. The effect was smaller than that produced by directly grounding the spacecraft to the chamber wall, but its repeatability under matched powered conditions shows that the available bonded conductive area contributed measurably to the spacecraft current balance. The additional dependence on spacecraft power state and probe orientation further indicates that the operating point was determined jointly by exposed conductive area, biased-probe collection and the available return-current paths [2].

This is directly relevant to fixed-bias mNLP operation because the commanded probe voltages are referenced to spacecraft ground rather than directly to the plasma. A negative spacecraft-potential shift therefore reduces the effective positive bias of every probe. For compact spacecraft, conductive exterior area should consequently be treated as an explicit plasma-instrument design variable rather than as an incidental property of the mechanical structure. The present experiment does not establish a universal minimum conductive area: 114.75 cm² is the quantified area difference between the two tested Munir-3 configurations, not a general requirement.

The OML analysis provides an additional boundary on interpretation. Only a small fraction of the grounded rotating measurements satisfied the adopted $R^2 \geq 0.95$ four-point linearity filter, and the retained fraction depended strongly on spacecraft orientation. Moreover, the active collector length was comparable to the estimated chamber Debye length. The reconstructed densities should therefore remain classified as OML-derived apparent electron densities rather than absolute chamber-density measurements; increased fit retention in Exp. 5.1 does not establish greater measurement accuracy [6].

The post-flight RF results provide a different but complementary system-level limitation. The communications chain exchanged valid packets over approximately 125–700 m but did not

establish a usable link at approximately 4.07 km under either tested modem configuration. Because final antenna gain, efficiency, matching, post-amplifier output and the complete receiver chain were not independently established, the tests cannot isolate the integrated antenna as the limiting element or reconstruct the flight link margin [2,5].

Similarly, the pre-flight EPS evidence establishes both successful integrated ground operation and an intermittent EPS-side overcurrent anomaly, but no flight electrical measurements exist. Together with the absence of independent spacecraft-release status, this prevents discrimination among deployment, activation, electrical, RF or other possible contributors to the missing telemetry. The post-flight results should therefore be interpreted as verified ground behaviours and design interactions, not as a reconstruction of the Munir-3 flight state [2].

For future implementations, spacecraft-level verification should combine final-configuration antenna characterization, simultaneous electrical loading, controlled conductive-area plasma tests and independent deployment-state observability. In particular, a fixed-bias plasma payload should be tested in its final electrically bonded exterior configuration, with spacecraft potential and effective probe operating margin evaluated under representative plasma conditions.

6. CONCLUSION

Munir-3 combined a frame-integrated antenna with a CC1200-based 436 MHz communication link and a four-channel fixed-bias mNLP in a 1U CubeSat using an electrically insulating GF30-PEEK structural architecture. No telemetry was received during flight, leaving spacecraft release, activation, RF operation and mNLP deployment unverified.

Post-flight RF testing demonstrated valid packet reception over approximately 125–700 m, while neither tested configuration established a usable link at approximately 4.07 km. Plasma-chamber testing indicated that restoring the flight exterior PCB panels, corresponding to 114.75 cm² of known ground-bonded copper relative to the reduced-area chamber configuration, made the

powered floating spacecraft 0.59–0.84 V less negative under matched conditions [2].

The principal engineering result is therefore a demonstrated coupling between dielectric structural/RF integration, spacecraft-grounded conductive area and fixed-bias plasma diagnostics. These quantities should be verified together at spacecraft level in the final as-built configuration. The available evidence does not establish a unique cause for the missing Munir-3 telemetry.

7. ACKNOWLEDGEMENT

The author gratefully acknowledges Andøya Space for its financial contribution to the launch campaign and for providing the launch site and necessary infrastructure, and NASA and NASA's Wallops Flight Facility for providing the launch vehicle, a rebuilt sounding rocket for scientific payloads, and for testing, payload integration and launch support. Special thanks go to Kolbjørn Blix, Co-PI of GHOST, Vice President of Andøya Space Sub-Orbital and leader of the Grand Challenge Initiative activities, and Christopher Koehler, Co-PI and leader of the RockSat-X programme, for enabling students to fly space hardware.

Above all, the author thanks all members of Lyra who carried out the work of developing Munir-3. With no prior spacecraft experience and an exceptionally short schedule, the team learned, designed, manufactured, integrated, tested and qualified a complete 1U CubeSat for flight. Their achievement was collective, built on persistence, trust and learning across unfamiliar disciplines, and shows what people can accomplish when they work together toward a goal beyond their initial experience. Lyra is a University of Oslo-based non-profit student organisation with Munir-3 students from the University of Oslo, Oslo Metropolitan University and the Norwegian University of Life Sciences.

The author gratefully acknowledges the University of Oslo, especially the Department of Physics, for workspace, facilities and encouragement. Within the 4DSpace research group, led by Wojciech Miloch, Lasse Clausen served as the author's MSc supervisor and adviser to Lyra's GHOST participation; Espen Trondsen guided development

of the mNLP front-end electronics and provided the flight probes later used in plasma-chamber testing; and Tore André Bekkeng's work on multi-needle Langmuir probes underpinned the payload. Torfinn Lindem is especially thanked for technical counsel, design reviews and hardware guidance, and for introducing the author and Lyra to the earlier CubeSTAR programme at UiO, which provided early inspiration and a foundation for the work.

The author further thanks Sigve S. Haugsnes of Andøya Space for engineering guidance and project reviews; Jonas S. Rignes and the UiO Instrumentation Laboratory for mechanical-design guidance and critical flight-readiness corrections; William N. Sigurdsson and the UiO Electronics Laboratory for PCB debugging and electronics support; and Sohail M. Mahmood for software debugging and subsystem integration. The author also thanks Anja Kohfeldt, Torbjørn Skauli, Ketil Røed, Sverre Holm and Helge Balk for advice and technical contributions.

Munir-3 depended on substantial support from industry, educational institutions and funding organisations. The author gratefully acknowledges Sparebankstiftelsen DNB, the project's principal financial supporter, together with Olav Thon Stiftelsen, Frifond, UiO Growth House, UNIK, EIDEL AS, Elprint AS, Integrated Detector Electronics AS, Norsk Elektro Optikk AS and Bredengen AS for financial and technical support. Special thanks go to Kongsberg Discovery and Kongsberg Defence & Aerospace for vibration testing; Altium for software and sponsorship; NCAB Group for PCB production and manufacturing guidance; Topro Elektronikk AS for assembling the flight PCBs under a demanding schedule; and IXYS AS, especially Sindre Steinsvik, for technical guidance, hardware support and exceptional availability.

The author further acknowledges Altitec for the flight battery package; Biesterfeld Group for materials and adhesive support for the antenna system; VINK Norway AS for supplying GF30 PEEK; the mechanical workshop at the Norwegian University of Life Sciences and the Norwegian Defence Research Establishment (FFI) for machining support; Finn Løken AS for POM material; Polyalkemi AS and Eugen Wiberger AB

for hardware support; and the Norwegian Radio Relay League for amateur-radio licence educational material. The Norwegian Space Agency is acknowledged for support of the project and future scientists and engineers, and ESA Academy for CubeSat training through the 2024 CubeSat Summer School.

Finally, the author acknowledges fellow GHOST participants from Clemson University, West Virginia State University, UiT Serpent, the University of Kentucky, the University of Puerto Rico Río Piedras and Andøya Upper Secondary School, with particular thanks to Andøya Upper Secondary School for launch-campaign participation and ground-station support.

REFERENCES

1. Bekkeng, T.A., Helgeby, E.S., Pedersen, A., Trondsen, E., Lindem, T. & Moen, J.I. (2019). Multi-Needle Langmuir Probe System for Electron Density Measurements and Active Spacecraft Potential Control on CubeSats. *IEEE Transactions on Aerospace and Electronic Systems* 55(6), 2951–2964.
2. Lyra (2025–2026). Munir-3 / GHOST Student Mission Reporting Document Package, internal project documentation package, Lyra, Oslo, Norway.
3. Mitsubishi Chemical Advanced Materials (2025). Ketron™ GF30 PEEK, product data sheet, issue/revision 15 April 2025, pp1–2.
4. Bekkeng, T.A., Jacobsen, K.S., Bekkeng, J.K., Pedersen, A., Lindem, T., Lebreton, J.-P. & Moen, J.I. (2010). *Design of a multi-needle Langmuir probe system. Measurement Science and Technology* 21, 085903.
5. Maral, G., Bousquet, M. & Sun, Z. (2020). *Satellite Communications Systems: Systems, Techniques and Technology*, 6th ed., John Wiley & Sons, Hoboken, NJ, USA, pp196–201.
6. Barjatya, A. & Merritt, W. (2018). Error analysis of multi-needle Langmuir probe measurement technique. *Review of Scientific Instruments* 89, 043507.